

Java Interview Notes

RAG-friendly interview preparation material covering core concepts and explanations.

1. Introduction to Java

Java is a high-level, object-oriented programming language designed to be portable across platforms. Java source code is compiled into bytecode that runs on the Java Virtual Machine.

2. JVM, JRE and JDK

JVM executes Java bytecode. JRE provides the JVM and libraries needed to run Java applications. JDK contains development tools such as the compiler along with the JRE.

3. Features of Java

Important Java features include object orientation, platform independence through bytecode, automatic memory management, exception handling, multithreading, and a large standard library.

4. Variables and Data Types

Java has primitive types such as int, long, float, double, char, boolean, byte, and short. Reference variables store references to objects.

5. Classes and Objects

A class is a blueprint containing fields and methods. An object is an instance of a class. Objects are created using the new operator.

6. Constructors

A constructor initializes an object when it is created. It has the same name as the class and has no return type. Constructors are not inherited by subclasses.

7. this and static

The this keyword refers to the current object. It can distinguish instance variables from parameters and can call another constructor. The static keyword makes a member belong to the class rather than to individual objects.

8. Inheritance

Inheritance allows a class to acquire accessible properties and behavior from another class. Java supports single, multilevel, and hierarchical class inheritance. Multiple inheritance of classes is not supported directly.

9. Polymorphism

Polymorphism means one interface or method name can represent different behavior. Method overloading is compile-time polymorphism, while method overriding supports runtime polymorphism.

10. Encapsulation

Encapsulation bundles data and methods inside a class and commonly uses private fields with public getter and setter methods to control access.

11. Abstraction

Abstraction focuses on essential behavior while hiding implementation details. Java provides abstraction through abstract classes and interfaces.

12. Interfaces and Abstract Classes

An interface defines a contract that implementing classes follow. An abstract class can contain abstract methods as well as implemented methods and instance state.

13. Exception Handling

Exceptions are handled using try, catch, finally, throw, and throws. try contains risky code, catch handles an exception, finally is used for cleanup, throw explicitly raises an exception, and throws declares possible exceptions.

14. Collections Framework

The Java Collections Framework provides reusable data structures and algorithms. ArrayList is a dynamic array, LinkedList is useful for linked-list operations, HashSet stores unique elements, and HashMap stores key-value pairs.

15. Multithreading

A thread is a lightweight unit of execution. Java supports multithreading through the Thread class, Runnable, and related concurrency utilities. Synchronization can help protect shared data.

16. Memory Management

Java uses automatic garbage collection to reclaim memory occupied by objects that are no longer reachable. Programmers do not normally explicitly free Java objects.